

RESEARCH ARTICLE

Thermal-Conductivity-Driven Dual-Wedge Thermoelectric Architecture for Intelligent Thermal Perception and Material Identification

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ABSTRACT

Maintaining a stable temperature difference within ultrathin thermoelectric sensors remains a key challenge for achieving skin-conformal and integrated thermal perception. Here, we present a dual-wedge thermoelectric (DWTE) architecture composed of two complementary wedge-shaped layers with distinct thermal conductivities. This structural configuration suppresses vertical heat diffusion while facilitating lateral heat flow, thereby establishing a stable lateral temperature gradient in thermoelectric films. This dual-wedge design not only enhances thermal response speed and signal stability but also significantly amplifies thermoelectric output under transient stimuli. Based on the thermal-conductivity-dominated thermal diffusion differences and an intelligent classification algorithm, the DWTE device integrated on an artificial hand enables rapid identification and classification of seven materials with an accuracy of 99.7%. Meanwhile, a 32-channel flexible thermoelectric array provides conformal real-time thermal mapping during object grasping. This work provides an effective solution to the long-standing trade-off between device thickness and thermal gradient stability, offering a universal strategy for fast-response thermoelectric sensing in advanced wearable and human-machine interactive systems.

1 | Introduction

Temperature sensors, as a key component of wearable electronics, have gained increasing attention in various applications such as health monitoring, soft robotics, and intelligent human-machine interaction [1–16]. Among various temperature sensing technologies, flexible thermoelectric sensors have emerged as promising candidates owing to their self-powered operation, simple device architecture, and capability for continuous and self-powered temperature monitoring [17–19]. Despite these

advantages, maintaining a stable temperature difference (ΔT) in ultrathin thermoelectric devices remains a persistent challenge. As device thickness decreases, vertical heat diffusion results in rapid thermal equilibration and weakened ΔT [20–23]. Various strategies have been proposed to mitigate this issue, including spiral or wave-shaped structures that elongate heat conduction paths and reduce lateral leakage [24, 25]. Although these strategies can maintain ΔT , they inevitably increase device thickness, compromising conformability and integration [26, 27]. Although materials engineering strategies, such as electrolyte regulation

[28], dopant design [29], and joint-free p-n integration [30], can enhance Seebeck coefficients, linearity, and response speed, these approaches primarily focus on thermoelectric conversion efficiency rather than the fundamental issue of ΔT preservation in ultrathin geometries [31–35]. Nevertheless, the intrinsic trade-off between device thickness and thermal gradient stability persists, as rapid thermal equilibration under ultrathin conditions inevitably leads to loss of effective temperature difference [36, 37]. Therefore, the core contradiction between the temperature difference and the ultrathin design remains unresolved.

In this work, we propose a dual-wedge thermoelectric (DWTE) architecture that breaks this constraint by regulating heat transport within ultrathin thermoelectric devices to establish a stable in-plane temperature gradient. The DWTE device integrates a high-thermal-conductivity liquid metal (LM) layer and a low-thermal-conductivity polydimethylsiloxane (PDMS) layer into complementary wedge-shaped geometries. The high-thermal-conductivity layer (top wedge) facilitates lateral heat transport, while the low-thermal-conductivity layer (bottom wedge) suppresses vertical heat diffusion, thereby promoting the establishment of a stable lateral thermal gradient. Such structural anisotropy not only enhances thermoelectric output and response speed but also stabilizes temperature gradients under transient stimuli. Finite element simulations and experimental validation confirm enhanced thermoelectric voltages compared to conventional thin-film architectures. The device enables rapid material recognition with 99.7% accuracy and real-time thermal mapping via a 32-channel conformal sensor array. This work resolves the critical trade-off between device thickness and temperature gradient, providing a universal strategy for fast-response thermoelectric sensing in next-generation wearable electronics.

2 | Results and Discussion

2.1 | Working Mechanism and Intelligent Material Identification of the DWTE devices

Establishing a stable ΔT while minimizing device thickness in thermoelectric sensors presents an inherent contradiction. Ionic thermoelectric materials typically require increased structural thickness to suppress heat diffusion and stabilize the ΔT (Figure 1a, i). Thin-film thermoelectric materials exhibit superior flexibility and conformability but struggle to maintain a vertical temperature difference, resulting in a weakened thermal gradient and reduced thermoelectric potential (Figure 1a ii). To address this limitation, we propose a dual-wedge thermoelectric (DWTE) architecture with complementary thermal properties, achieving stable lateral ΔT and thermoelectric output voltage in ultrathin configurations (Figure 1a iii). This dual-wedge design also facilitates rapid response to thermal stimuli, ensuring both efficient heat conduction and high thermoelectric output voltage. The high-thermal-conductivity liquid metal (LM) layer efficiently promotes vertical heat conduction, while the low-thermal-conductivity PDMS layer suppresses vertical heat diffusion, establishing an in-plane thermal gradient within the thermoelectric film (Figure 1b). The lateral ΔT facilitates the migration of charge carriers from the high-temperature region to the low-temperature region, generating a stable thermoelectric output voltage (ΔV). This structure also facilitates rapid thermal

response and stable thermoelectric potential, ensuring efficient response under transient stimuli.

The dual-wedge working mechanism further reveals that the ΔT is dominated by thermal conductivity (k). Materials with higher k enable faster heat propagation and higher ΔT , whereas lower- k materials exhibit slower heat transfer and lower ΔT . To quantify this correlation, a dimensionless slope index ($S = \Delta V_{\max}/t_k$) is introduced that combines the peak thermoelectric voltage ($\Delta V_{\max}$) and thermal response time (t_k) [38–40]. As shown in Figure 1c, serving as a unique material signature, the slope index S facilitates highly accurate classification through artificial neural networks. When used as input features for an artificial neural network, the slope index S enables accurate classification of materials based on their intrinsic thermal conductivity, achieving material identification.

2.2 | Thickness-Mediated Thermal Gradient and Thermoelectric Response

The lateral ΔT is crucial for achieving reliable thermoelectric responses under transient thermal stimulation. The effect of dual-wedge thickness on thermal gradient and thermoelectric response was systematically investigated using finite element simulations and experimental measurements. Both the simulation results and infrared thermal imaging clearly reveal pronounced thickness-dependent variations in lateral heat distribution (Figure 2a,b). In ultrathin devices (200 μm), limited thermal capacity and poor vertical insulation result in rapid vertical heat diffusion and make it difficult to establish and sustain a stable lateral ΔT . In contrast, excessively thick devices (2000 μm) exhibit high thermal resistance, which hinders heat injection at the hot side. Both the restricted heat input and the slow in-plane diffusion contribute to the reduced lateral ΔT . In addition, the nearly identical temperature distributions with and without the thermoelectric layer indicate that, owing to its ultrathin nature, thermal diffusion within the thermoelectric film has a negligible impact on the temperature difference and overall thermoelectric performance governed by the dual-wedge architecture (Figure S1).

An optimal balance is achieved at an intermediate thickness of 500 μm , where these competing thermal effects are effectively balanced. At an optimized thickness, the high-thermal-conductivity layer facilitates rapid heat delivery to the hot side of the thermoelectric film, whereas the low-conductivity layer serves as an effective thermal barrier that minimizes vertical dissipation and preserves the cold side at a lower temperature, thereby sustaining a stable temperature gradient across the device. This configuration yields the largest and most stable lateral ΔT between the hot and cold regions (Figure 2c,d). Both simulated and experimentally extracted results (Figure 2e) consistently demonstrate that the DWTE device based on 500 μm dual-wedge exhibits the highest lateral ΔT , which directly translates to the maximum transient thermoelectric output voltage observed in experiments (Figure 2f). Meanwhile, the stacking sequence of the two wedges plays a critical role in establishing an effective in-plane temperature difference. The PDMS-top configuration significantly increases thermal resistance and limits vertical heat transport, thereby reducing both the thermal response and

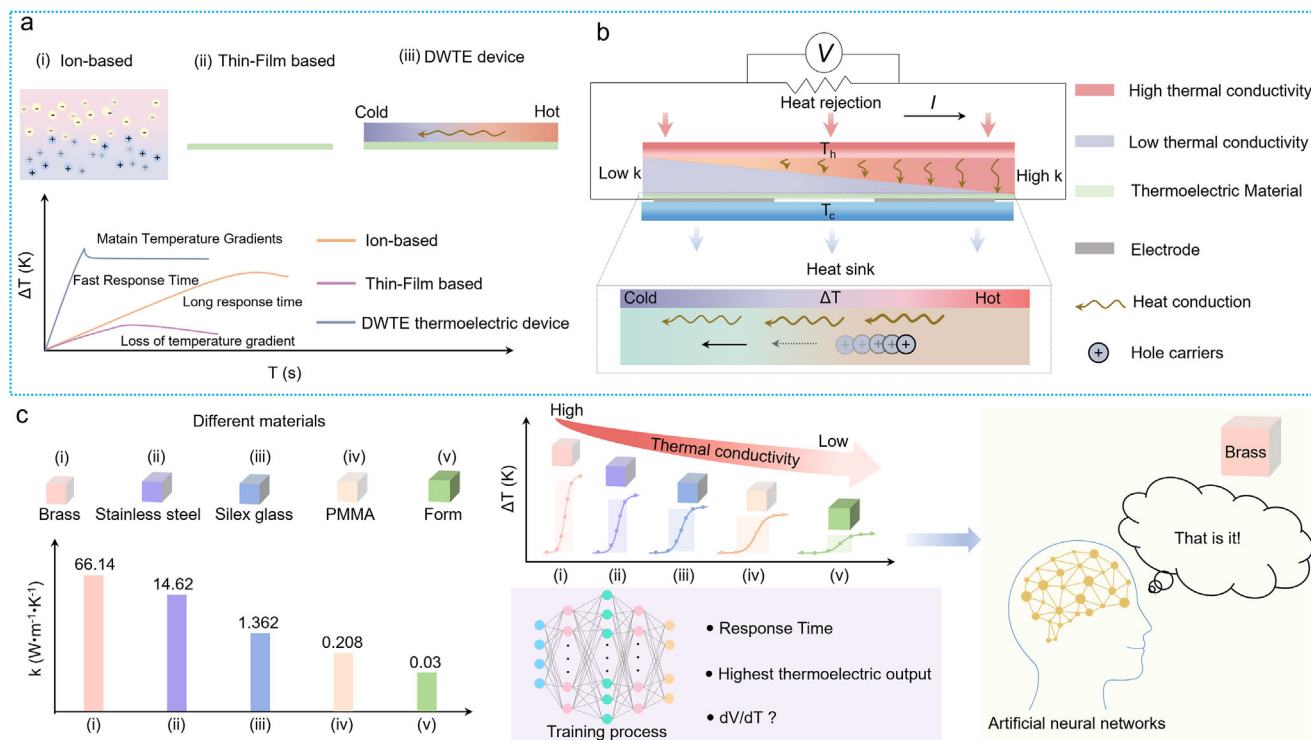


FIGURE 1 | Working mechanism and intelligent material identification of the DWTE devices. (a) Comparative illustration of ΔT formation in three types of thermoelectric devices, including ionic-based, thin-film-based, and DWTE devices. (b) Schematic of the DWTE working principle and the formation of the lateral thermal gradient. (c) Schematic illustration of thermal conductivity-based material identification.

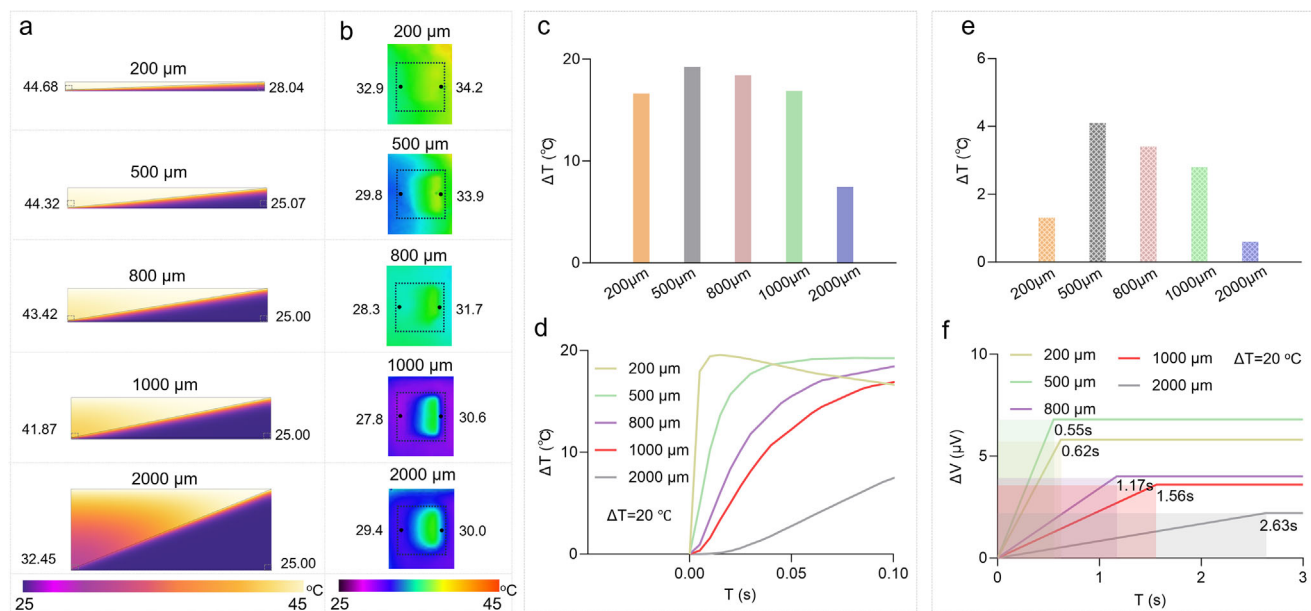


FIGURE 2 | Thickness-mediated thermal gradient and thermoelectric response. (a) Finite element simulations of transient heat transfer in DWTE devices with various dual-wedge thicknesses (200, 500, 800, 1000, and 2000 μm) under a 20°C thermal stimulus. (b) Top-view infrared thermal images showing the lateral temperature distribution across DWTE structures with different thicknesses. (c) Simulated transient ΔT values for DWTE devices with varying thicknesses under a 20°C stimulus. (d) Variation in lateral ΔT , the temperature difference between the average temperatures at the two ends of the thermoelectric film in the simulation model (Figure 2a). (e) Experimentally measured ΔT value between the hot and cold sides under a 20°C stimulus, captured by infrared thermal imaging. (f) Measured time-dependent thermoelectric output voltages of DWTE devices with different thicknesses under a 20°C stimulus.

the generation of an effective lateral temperature difference (Figures S2–S4).

Under steady-state conditions, thicker DWTE devices (2000 μm) can accumulate heat at the hot side, producing larger ΔT and higher output voltages due to enhanced vertical insulation and asymmetry in heat propagation paths (Figure S5). However, the increased thickness results in a prolonged thermal diffusion path and equilibrium time, leading to sluggish response dynamics, which are unfavorable for real-time sensing. In contrast, the 500 μm DWTE device achieves an optimal trade-off between transient response speed, sustained ΔT , and output voltage, making it highly suitable for real-time and dynamic thermoelectric sensing. The thickness-mediated thermal gradient and thermoelectric response based on the DWTE design provide critical design guidance for ultrathin, fast-response thermal perception systems.

2.3 | Structural Optimization and Material Universality of DWTE Devices

Four representative structural configurations were fabricated to validate the structural superiority of the DWTE design (Figure 3a). All devices possessed the same total thickness (1 mm) and identical material compositions, ensuring that any variations in performance originated solely from structural configuration. The cold side was maintained at 25°C and the hot side gradually heated to 50°C. The DWTE architecture (Structure A) exhibited the highest thermoelectric output voltage compared to the other three configurations (Figure 3b). The enhanced performance of Structure A arises from its uniform temperature gradient across the thermoelectric film caused by the DWTE design. High-conductivity LM layer rapidly conducts the heat toward the hot region, while the low-conductivity PDMS layer effectively blocks vertical heat conduction, enabling the formation of a self-sustained in-plane thermal gradient. In contrast, Structure B (side-by-side LM and PDMS) suffers from limited lateral ΔT due to separated conduction domains. The superior performance of the DWTE structure originates from the formation of a continuous in-plane temperature gradient along the thermoelectric film. The overall thermoelectric voltage arises from the integrated temperature difference across the entire device length. However, the side-by-side configuration (Structure B) produces a localized, step-like temperature difference that limits effective voltage generation (Figures S6 and S7). Structure C, composed entirely of LM, exhibits rapid thermal equilibration due to its high thermal conductivity that suppresses ΔT formation. Meanwhile, Structure D, made of thermally insulating PDMS, confines heat within the top surface, resulting in poor vertical heat conduction. These results confirm that the complementary DWTE design in Structure A provides the optimal pathway for maintaining a stable and high ΔT .

The consistent thermoelectric outputs obtained from multiple devices confirm good batch-level fabrication reproducibility, highlighting the robustness of the DWTE device manufacturing process (Figure S8). In addition, the DWTE device exhibits a dimension-dependent thermoelectric output voltage due to the vertical heat conduction within the DWTE structure and edge heat dissipation (Figure 3c). In smaller devices, the reduced cross-section of the PDMS insulation limits vertical heat conduction,

while the hot side rapidly accumulates heat, establishing a higher temperature gradient and output voltage. Enlarging the area increases the edge exposure and heat dissipation, lowering the gradient and output voltage. An active area of $5 \times 5 \text{ mm}^2$ was identified as the optimal configuration. Therefore, optimizing the vertical insulation and controlling edge thermal losses are key to achieving a high and stable temperature difference within ultrathin DWTE systems.

To evaluate the practical applicability of the optimized DWTE structure in bio-integrated thermal sensing, the DWTE device was attached onto human skin for perceiving external ambient temperature changes, where one side of the device maintains body-temperature equilibrium (35°C), while the opposite side was exposed to external stimuli ranging from 0°C to 70°C (Figure 3d). As shown in Figure 3e, the DWTE device exhibited distinct bidirectional and linear thermoelectric sensitivity, demonstrating positive voltages in response to stimuli warmer than 35°C and negative voltages for cooler stimuli. Meanwhile, the 500 μm DWTE device achieves an optimal output voltage. This linear and symmetric thermal response allows the device to detect ambient temperature changes, indicating its potential for intelligent thermal perception in wearable bioelectronics.

The universality of the DWTE architecture was also validated across various thermoelectric materials, including gelatin gel, polyacrylamide (PAM), and PEDOT:PSS (Figure 3f–h). These devices exhibit consistent trends in output voltage with an optimized thickness of 500 μm . It confirms that the formation of in-plane thermal gradient is primarily dominated by the DWTE architecture instead of the thermoelectric material, suggesting the adaptability of the DWTE concept in soft and wearable thermoelectric sensors. In addition, beyond thermal sensing, the dual-wedge architecture is also applicable to other energy conversion scenarios, as it enables stable temperature-gradient regulation for both thermoelectric power generation and photothermal-induced electrical conversion (Figures S9 and S10).

2.4 | Thermoelectric Performance and Stability of DWTE Devices

To further evaluate the voltage stability of the DWTE device, retention tests were conducted under a constant thermal gradient ($\Delta T = 5^\circ\text{C}$). As shown in Figure 4a, DWTE devices maintain higher voltage retention than pristine PEDOT:PSS films, owing to the stable lateral ΔT induced by the dual-wedge architecture. This confirms that the DWTE architecture allows for durable and stable thermoelectric output even under steady-state conditions.

A skin-referenced bidirectional thermoelectric response was subsequently demonstrated (Figure 4b). The DWTE sensor exhibits positive voltage outputs during heating ($T > 35^\circ\text{C}$) and negative outputs during cooling ($T < 35^\circ\text{C}$). To further investigate the sensor's adaptability in complex environments, stepwise heating and cooling experiments (35°C–70°C, $T_{\text{ref}} = 35^\circ\text{C}$) were conducted. Moreover, when tested on a heating stage under ambient room-temperature conditions, the DWTE device shows stable temperature sensing from 0°C to 75°C with excellent linearity ($R^2 = 0.9989$) and no discernible discontinuity across the LM melting point ($\sim 16^\circ\text{C}$) (Figures S11 and S12). It indicates a reversible

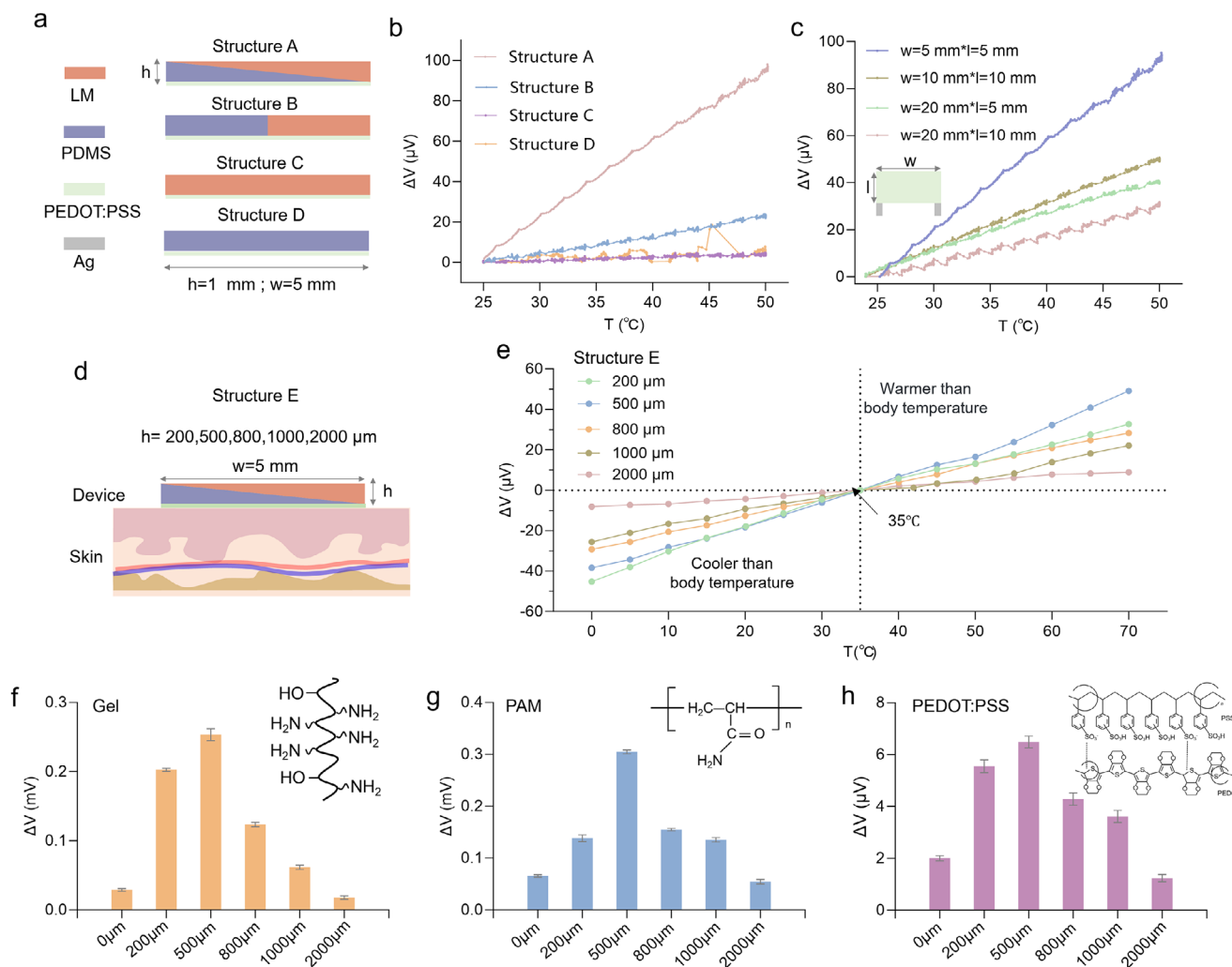


FIGURE 3 | Structural optimization and material universality of DWTE devices. (a) Schematic illustration of four structural configurations used. (b) Thermoelectric output voltage of the four structures with the cold side fixed at 25°C and the hot side increased from 25°C to 50°C . (c) Optimization of thermoelectric film area under identical thermal conditions. (d) Schematic illustration of a DWTE device attached onto human skin for perceiving external temperature stimuli, where one side of the device maintains body-temperature equilibrium (35°C). (e) Thermoelectric output voltages of DWTE devices with different thicknesses in response to thermal stimuli below and above body temperature, demonstrating linear thermal sensing behavior. Validation of the DWTE configuration across different thermoelectric materials (f) gelatin gel, (g) polyacrylamide (PAM), and (h) PEDOT:PSS.

ΔV - T response without significant lag and hysteresis (Figure 4c). The transient response of the DWTE sensor was further investigated under different temperature gradients ($\Delta T = 5^{\circ}\text{C}$ – 45°C). As shown in Figure 4d, the DWTE device exhibits increasing response and recovery time with increasing temperature stimulus due to the intrinsic heat transfer dynamics. It should be noted that the contact thermal resistance at the interfaces between the wedge-shaped layers and the thermoelectric film cannot be negligible in practical devices, which contributes to the slower thermal response compared to idealized simulations (Figure 2d). This further suggests that reducing interfacial thermal resistance could enable additional improvements in temperature response speed. Compared with previously reported thermoelectric sensors (Figure 4e), the DWTE device demonstrates rapid response and recovery time, indicating rapid adaptability to fluctuating thermal stimulus [41, 42, 7, 43–46]. Meanwhile, the DWTE device exhibited clearly distinguishable and repeatable output signals under a small temperature stimulus of 0.1°C across different temperature ranges, indicating the capability to detect

small temperature stimuli across different operating temperature ranges (Figure 4f).

The mechanical flexibility of the DWTE device was also investigated. Photographs of the fabricated DWTE device with sub-millimeter thickness from multiple perspectives are provided in Figure S13. Optical images in Figure 4g show that the device maintains its structural integrity under repeated bending (80° , 140°) and recovers to its initial state. As presented in Figure 4h, voltage outputs recorded under a fixed $\Delta T = 15^{\circ}\text{C}$ remain nearly identical across the initial, bent (90°), and recovered states, indicating robust and stable thermoelectric behavior under mechanical deformation. In addition, the DWTE sensor's resistance to pressure interference was examined. Figure 4i presents the thermoelectric outputs measured at applied pressures of 0, 10, 20, 30, and 40 kPa, and the nearly unchanged voltage outputs confirm that the DWTE sensor is pressure-insensitive, highlighting the robustness of the dual-wedge design and ensuring reliable operation in contact-based or conformal configurations.

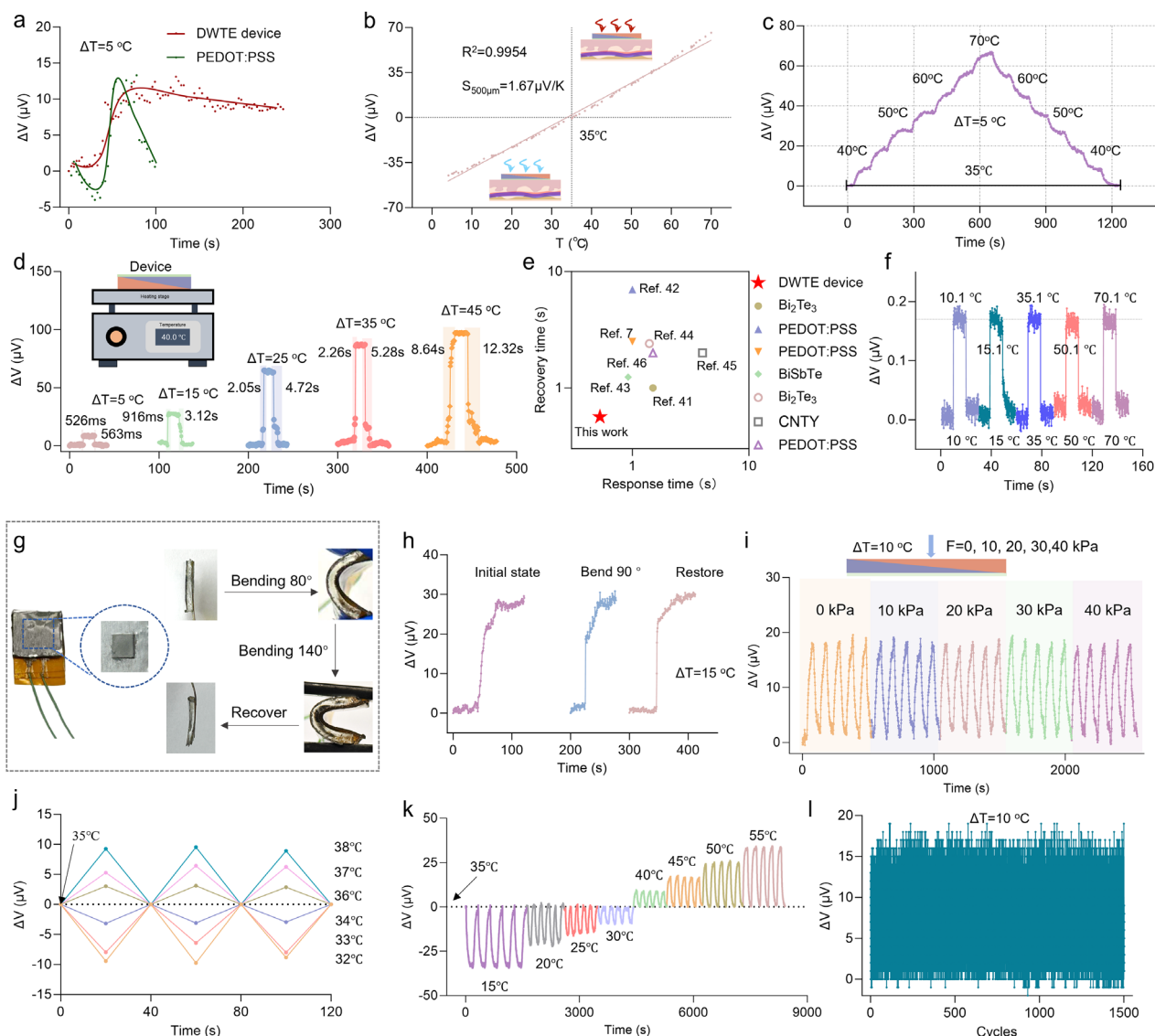


FIGURE 4 | Thermoelectric performance and stability of DWTE devices. (a) Retention of the output voltage over time under constant temperature gradient ($\Delta T = 5^\circ\text{C}$) for DWTE devices and PEDOT:PSS thermoelectric films. (b) Skin-referenced bidirectional thermal response, showing voltage increase under heating ($T > 35^\circ\text{C}$) and decrease under cooling ($T < 35^\circ\text{C}$). (c) Relative voltage variation under stepwise temperature changes from 35°C to 70°C , $T_{\text{ref}}=35^\circ\text{C}$. (d) Response and recovery times of DWTE device under various ΔT (5°C , 15°C , 25°C , 35°C , and 45°C). (e) Comparison of response and recovery times with previously reported thermoelectric devices. (f) Thermoelectric response to minimal stimulus of the DWTE device under different temperatures, indicating repeatable output signals across different temperature ranges. (g) Optical images of the DWTE device under various bending states. (h) Output voltage of the device in the initial, bent (90°), and recovered states ($\Delta T = 15^\circ\text{C}$). (i) Voltage outputs under different applied pressures (0–40 kPa), demonstrating pressure-insensitive thermoelectric response. (j) Cyclic stability during continuous thermal cycling between 32°C and 38°C , $T_{\text{ref}} = 35^\circ\text{C}$. (k) Long-term repeatability of thermoelectric response under various thermal stimuli between 15°C and 55°C , $T_{\text{ref}} = 35^\circ\text{C}$. (l) Cycling stability of the DWTE device under repeated thermal stimuli ($\Delta T = 104^\circ\text{C}$).

Long-term cyclic and repeatability tests further validate the stability of the DWTE design. During continuous thermal cycling between 32°C and 38°C ($T_{\text{ref}} = 35^\circ\text{C}$), the device exhibits repeatable output voltages (Figure 4j). The DWTE device exhibits reproducible bidirectional voltage outputs during cyclic temperature variations between 15°C and 55°C ($T_{\text{ref}} = 35^\circ\text{C}$), exhibiting positive and negative signals during heating and cooling, respectively (Figure 4k). The nearly identical amplitude in output voltages across cycles confirm excellent repeatability of the device. In addition, the DWTE device maintains stable output without noticeable degradation or drift over 1500 cycles, demonstrating

robust operational stability (Figure 4l). These results confirm that the DWTE design enables a robust thermoelectric platform for reliable perception in wearable electronics [20, 47–49].

2.5 | Tactile Material Identification Enabled by DWTE-Based Thermoelectric Sensing and Machine Learning

To endow robotic tactile systems with human-like material recognition capabilities, seven representative materials spanning over

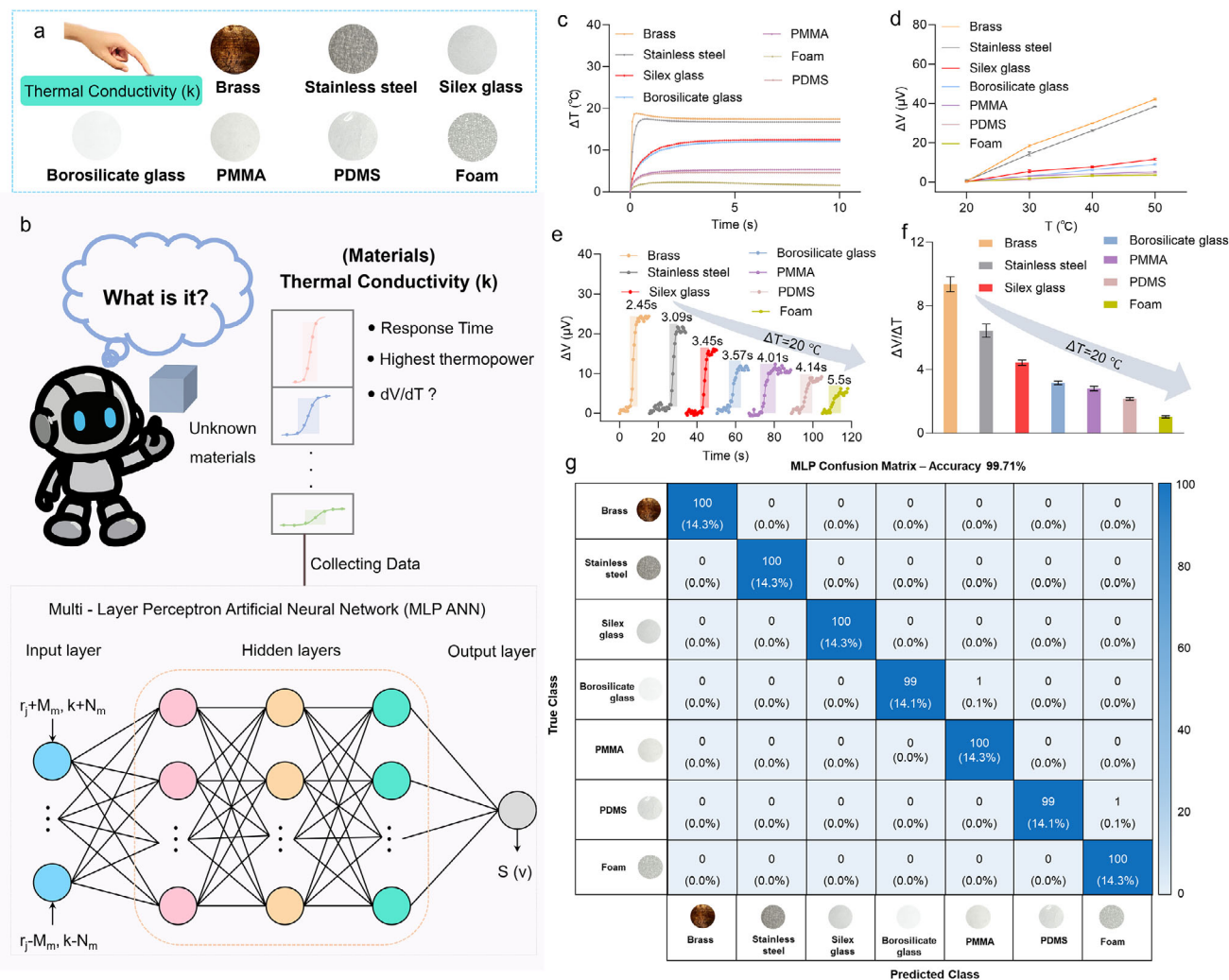


FIGURE 5 | Tactile material identification enabled by DWTE-based thermoelectric sensing and machine learning. (a) Optical images of seven test materials with distinct thermal conductivity. (b) Schematic of the material identification process, including measurement of thermoelectric output (ΔV), response time (t_k), and $\Delta V/t_k$ values. (c) Finite element simulations of ΔT generation under 20°C thermal stimulus for materials with different k . (d) Measured ΔV as a function of ΔT (0°C – 30°C). (e) Measured ΔV and t_k values ($\Delta T = 20^\circ\text{C}$). (f) Extracted slope parameter S ($\Delta V/t_k$) as an indicator for materials identification. (g) multilayer perceptron (MLP) neural network trained on S values.

three orders of magnitude in thermal conductivity (k), including brass, stainless steel, silex glass, borosilicate glass, PMMA, PDMS, and foam were selected (Figure 5a). The thermoelectric output voltage (ΔV) and the thermal response time (t_k) upon contact with each material were measured using the DWTE device. The slope parameter $S = \Delta V/t_k$ was extracted as a comprehensive feature representing both the magnitude and dynamic of thermoelectric conversion. This parameter combines the magnitude and speed of the thermoelectric response under thermal stimulation, providing a robust metric for material discrimination (Figure 5b).

Finite element simulations indicated that the ΔT values are critically determined by the k values of materials. High- k materials (e.g., brass, $k \approx 66.14 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$) exhibit rapid heat conduction via efficient electron-mediated lattice conduction, achieving $\Delta T \approx 17.43^\circ\text{C}$ under a 20°C stimulus. In contrast, low- k materials (e.g., foam, $k \approx 0.03 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$) exhibit significant thermal localization owing to the porous, air-trapping microstructures, resulting in minimal ΔT (Figure 5c). The thermoelectric potential under $\Delta T = 0^\circ\text{C}$ – 30°C further demonstrated that metals with

high k exhibit steep voltage increases, driven by rapid charge-carrier diffusion under strong temperature gradients, whereas low- k polymers display slower, weaker responses (Figure 5d). At a fixed $\Delta T = 20^\circ\text{C}$, brass achieved $\Delta V = 24.4 \mu\text{V}$ with $t_k = 2.45 \text{ s}$, while foam produced only $\Delta V = 6.2 \mu\text{V}$ with $t_k = 5.5 \text{ s}$ (Figure 5e).

The slope parameter S combines the two correlated factors, where the ΔV increases with ΔT , and the t_k decreases with k , making S an indicator of the intrinsic thermal behavior of materials (Figure 5f). Inputting S into a multilayer perceptron (MLP) classifier enables classification of the seven materials with a high accuracy of 99.71%. However, when using the thermoelectric voltage amplitude ΔV alone as the input feature, the classification accuracy is 91.43% (Figure S14). Using the thermal response time t_k alone yields a lower accuracy of 74.29% (Figure S15), indicating limited discriminative capability. Combining ΔV and t_k into a 2D feature vector improves the accuracy to 94.29% (Figure S16), demonstrating that temporal dynamics provide complementary information to the voltage magnitude. As shown in Figure 5g, the confusion matrix indicates excellent classification accuracy, with

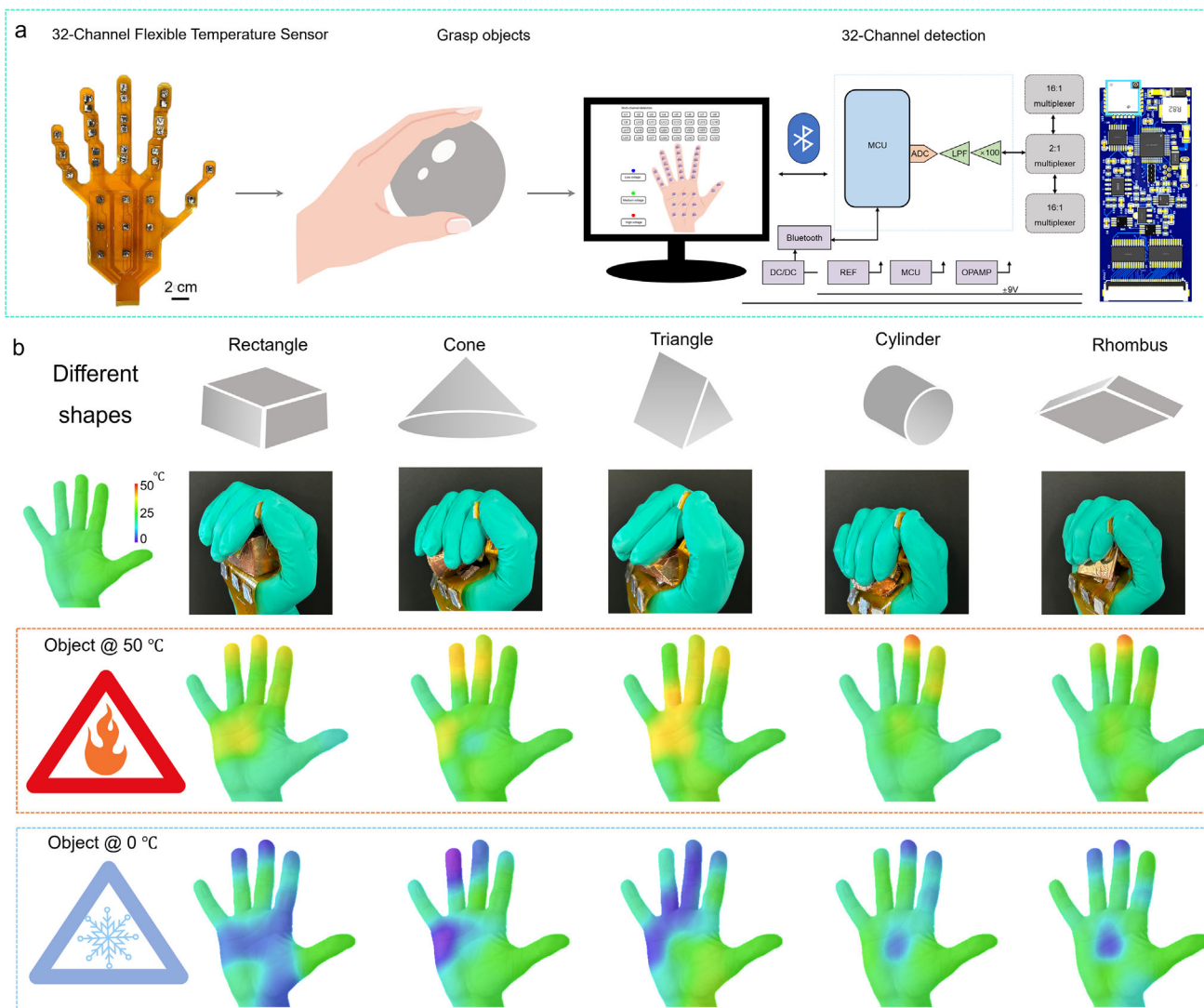


FIGURE 6 | Real-time shape and temperature mapping using a 32-channel DWTE-based thermoelectric sensor array. (a) Schematic of the integrated sensing system, including the 32-channel sensor array, and a three-level multiplexing circuit for data acquisition across all channels. (b) Demonstration of simultaneous shape and temperature recognition. Five geometric objects (cube, cone, triangle, cylinder, and rhombus) preconditioned to 0 °C or 50 °C generate distinct voltage reflecting both object contours and thermal states.

precise differentiation between all materials, including similar metals and low- k polymers. By encoding material-dependent heat transfer into measurable electrical features (ΔV , t_k , S), the DWTE device achieves precise material recognition with high tactile accuracy, indicating its potential for intelligent tactile perception [33, 50, 51, 26, 52].

2.6 | Real-Time Shape and Temperature Mapping Using a 32-Channel DWTE-Based Thermoelectric Sensor Array

To demonstrate the practical potential of the DWTE design, a 32-channel thermoelectric sensor array was further exploited into a flexible, hand-shaped platform for real-time tactile-thermal perception (Figure 6a; Movie S1). Each sensing unit was based on a DWTE element, allowing local temperature-difference generation and independent signal output. The array was connected to a three-level multiplexing circuit for sequential time-division

signal acquisition across all channels. The analog thermoelectric signals were amplified, filtered, digitized, and wirelessly transmitted to a computer through a microcontroller (MCU) for real-time visualization and processing. The system achieves low-noise, high-density signal collection while maintaining mechanical flexibility and conformity.

The integrated system demonstrated simultaneous shape and temperature mapping. As shown in Figure 6b, when the artificial hand grasped objects with different geometric shapes (cube, cone, triangular prism, cylinder, and rhombus), each produced a distinct thermal pattern corresponding to the contour and contact area of the object. Moreover, objects preconditioned at 0 °C and 50 °C generated clearly distinguishable thermal maps, revealing the spatial temperature distribution across the hand surface. Warm objects (>25 °C) produced positive signal shifts (red), while cold objects (<25 °C) induced negative shifts (blue). The same objects exhibit consistent spatial patterns at different temperatures. The system could therefore differentiate both

geometric form and thermal state in real time. These results confirm that combining DWTE devices with multiplexed circuit integration enables high-resolution, dynamic thermal imaging. It provides a promising platform for advancing artificial-skin, prosthetic, and human-machine interface applications, where real-time perception of shape and temperature is essential for intelligent interaction.

3 | Conclusion

In summary, we have developed a dual-wedge thermoelectric (DWTE) architecture to address the long-standing trade-off between device thickness and temperature-difference in flexible thermoelectric sensors. By integrating high and low-thermal-conductivity materials into complementary wedge geometries, the DWTE design enables effective heat regulation, establishing a stable in-plane temperature gradient within sub-millimeter-thick devices. The simulated and experimental results confirm that the DWTE architecture contributes to enhanced temperature gradients and voltage output compared with a conventional design. When coupled with intelligent algorithms, the tactile system enables precise material identification with an accuracy of 99.7%. A 32-channel sensor array achieves real-time mapping of shape and temperature. The proposed DWTE architecture provides a universal route for high-performance thermal perception of ultrathin thermoelectric devices.

4 | Experimental Section

4.1 | Materials and Reagents

Gelatin (250 g Bloom) was obtained from Aladdin, Shanghai, China. $K_4Fe(CN)_6 \cdot 3H_2O$; $K_3Fe(CN)_6$; Acrylamide; N,N'-Methylenebis was obtained from Macklin, Shanghai, China. $K_2S_2O_8$ was obtained from Sinopharm Chemical Reagent Co., Ltd. PDMS Sylgard 184 was obtained from Dow Corning. The liquid metal (LM; GaIn) was obtained from Dongguan Hua Titanium Material Technology Co. Ltd. PEDOT:PSS (PY200), Ag (LY-50) was obtained from Shenzhen Yilai Technology Co., Ltd.

4.2 | Preparation of DWTE Devices

All DWTE devices were fabricated through a standardized process involving mold formation, electrode and film deposition, and structural assembly (Figure S17). First, silver electrodes ($2 \times 7 \text{ mm}^2$) were patterned onto polyimide (PI) substrates using a flexible electronics printer. The Ag ink was dispensed through a 0.25 mm needle at a pressure of 200 kPa and a speed of $1 \text{ mm} \cdot \text{s}^{-1}$, followed by curing at 80°C for 30 min. Thermoelectric films composed of PEDOT:PSS were then printed atop the electrodes using a 0.21 mm needle at 120 kPa and the same dispensing speed, forming films of various dimensions (5×5 , 10×10 , 20×5 , and $20 \times 10 \text{ mm}^2$). After printing, the assemblies were again cured at 80°C for 30 min, and silver-plated copper wires were attached to the electrode terminals. Second, custom molds for the PDMS substrate were fabricated via 3D printing using photosensitive resin. The PDMS base and curing agent were mixed at a 10:1 mass ratio, degassed under vacuum, poured

into the molds, and cured at 80°C for 2 h to form dual-wedge architecture substrates. The PDMS wedge structure was sealed with aluminum tape to create an enclosed cavity, and liquid metal (Ga:In = 75:25) was injected into the cavity. Subsequently laminated onto the PI/Ag/PEDOT:PSS layer to form the complete device. To ensure robust interfacial bonding, the contact surfaces of the PDMS wedge substrate and the PI substrate were treated with O_2 plasma immediately prior to lamination (60 s). The two layers were then brought into conformal contact, gently pressed to remove trapped air. Depending on the structure, different thermally conductive materials were injected. For structures A and E, LM (GaIn, melting point $\approx 16^\circ \text{C}$) was injected into wedge-shaped cavities with thicknesses ranging from 200 to 2000 μm . For structure B, PDMS and LM were filled into separate left and right compartments (each $2.5 \times 5 \text{ mm}^2$). For structures C and D, the entire $5 \times 5 \text{ mm}^2$ cavity was uniformly filled with either LM or PDMS.

4.3 | Preparation of Thermoelectric Materials

Laser-patterned poly (methyl methacrylate) (PMMA) substrates (2 mm thick) were fabricated using a KBBA K3020 laser engraving system. Each substrate featured a $10 \times 10 \text{ mm}^2$ outer frame with a central $5 \times 5 \text{ mm}^2$ cavity and lateral channels for electrode placement. Carbon paper electrodes ($2 \times 5 \text{ mm}^2$) were affixed to opposing sides of the cavity using cyanoacrylate adhesive (502), and nylon-silver wires were bonded to the electrodes with silver paste. The central cavity was reserved for subsequent hydrogel infusion. Two types of thermoelectric hydrogels were prepared. For the gelatin-based system, 3 g of gelatin, 0.1365 g of potassium ferrocyanide trihydrate ($K_4Fe(CN)_6 \cdot 3H_2O$), and 0.1588 g of potassium ferricyanide ($K_3Fe(CN)_6$) were dissolved in 8 mL of deionized water and magnetically stirred at 1000 rpm for 5 min. The resulting viscous solution was immediately cast into the cavity of the PMMA assembly and sealed with medical adhesive tape. For the polyacrylamide (PAM) system, 1.25 g of acrylamide monomer was dissolved in 4 mL of deionized water, followed by the addition of 25 mg of potassium persulfate ($K_2S_2O_8$) as the initiator and 7.5 mg of N,N'-methylenebisacrylamide (MBAA) as the crosslinker. After thorough mixing, the solution was poured into a separate PMMA mold and thermally cured at 60°C for 1 h. The solidified PAM hydrogel block was then transferred into the cavity of the PMMA electrode assembly and sealed using medical tape. Finally, for both types of thermoelectric hydrogels, a prefabricated LM/PDMS wedge structure ($5 \times 5 \text{ mm}^2$, 500 μm thick) was carefully aligned and laminated atop the hydrogel-filled substrate to form the complete device.

4.4 | Preparation of 32-Channel DWTE-Based Thermoelectric Sensor Array

The 32-channel DWTE-based thermoelectric sensor array was designed in electronic design automation (EDA) software, from which Gerber files were exported for printed circuit board (PCB) fabrication. As shown in Figure S18, the resulting array consists of 32 pairs of silver electrodes (each wire: $2 \times 5 \text{ mm}^2$) patterned on a polyimide (PI) substrate, forming independent sensing units arranged in a bionic hand structure. Thermoelectric sensing films

composed of PEDOT:PSS ($5 \times 5 \text{ mm}^2$) were deposited on each electrode pair using a flexible electronics printer. The solution was dispensed through a 0.21 mm needle at a pressure of 120 kPa and a speed of $1 \text{ mm} \cdot \text{s}^{-1}$, followed by thermal curing at 80°C for 30 min. Subsequently, prefabricated LM/PDMS wedge structures ($5 \times 5 \text{ mm}^2$, 500 μm thick) were carefully aligned and bonded atop each PEDOT:PSS film to complete the assembly of the 32-channel DWTE-based thermoelectric sensor array. Raw signal data were imported into MATLAB and preprocessed to ensure suitability for supervised learning (Figure S19).

4.5 | Thermal Visualization Methodology for Palmar Temperature Distribution

The signal acquisition system was powered by a 3.7 V lithium battery, with voltage levels regulated via a DC-DC conversion module to support all circuit components. The 32-channel sensor array was interfaced with a custom-designed PCB and connected to the data acquisition module for analog signal routing and processing. (Figure S20). To enable intuitive observation of thermal data, a real-time palm temperature visualization system was developed based on the 32-channel DWTE-based thermoelectric sensor array. Core temperature data were transmitted via Bluetooth and parsed on a host computer, where they were dynamically rendered into heatmaps using a customized visualization algorithm. Each set of sensor readings was first normalized and mapped to the HSV color space, with temperature values translated into a continuous hue scale ranging from blue-purple (cold) to red (hot). To simulate lateral heat conduction across the palm, a 2D Gaussian diffusion algorithm was implemented, allowing each temperature point to influence not only its own coordinates but also neighboring pixels. This yielded a smoothed spatial distribution of thermal intensity. A palm-shaped mask was applied to restrict rendering to the anatomical sensing region. To enhance perception of physiologically relevant regions, the system incorporated an adaptive contrast enhancement module. Through statistical analysis, the temperature range with the highest density of data points was identified and subjected to nonlinear remapping and gamma correction. This process improved color resolution and local contrast, making subtle thermal variations more distinguishable. Temporal coherence was achieved via interframe interpolation. By linearly interpolating between consecutive Bluetooth frames, intermediate images were generated to construct a smooth animation, eliminating visual discontinuities and improving user experience (Figure S21). All volunteers signed informed consent before the experiments according to the guidelines of the Institutional Research Ethics Board.

4.6 | Characterization and Measurements

Temperature-controlled signal acquisition was conducted using custom-designed hardware and software platforms. Conductive ink printing for the fabrication of sensor electrodes was carried out using the Prtronic Scientific 4 system (Shanghai Zhongbin Technology Co., Ltd.). Infrared thermal images were captured using a Hikmicro P20MaxV2 infrared camera. Finite element simulations were performed using COMSOL Multiphysics 5.4 to model temperature distributions.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File 1: adfm73984-sup-0001-SuppMat.docx.

Supporting File 2: adfm73984-sup-0002-MovieS1.mp4.